

The fragments of the mathematical histories of EUDEMUS of Rhodes

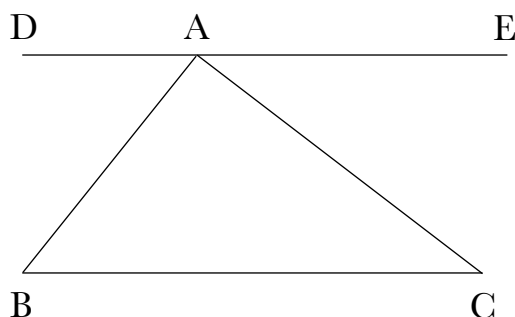
History of Geometry

[Thomas, *Selections illustrating the history of Greek mathematics*, vol. 1, p. 144-155] (*fr. 133*)

Eudemus in his history of geometry attributes the theorem itself [*Elements* 1.26] to Thales, saying that the method by which he is reported to have determined the distance of ships at sea shows that he must have used it. (*fr. 134*)

It [*Elements* 1.15] was first discovered by Thales, Eudemus says, ... (*fr. 135*)

Eudemus the Peripatetic attributes to the Pythagoreans the discovery of this theorem, that every triangle has internal angles equal to two right angles, and says they demonstrated it as follows. Let ABC be a triangle, and through A draw a line DE parallel to BC. Then since BC and DE are parallel, the alternate angles are equal, and angle DAB is therefore equal to ABC and EAC to ACB. Add the common angle BAC. Then angles DAB, BAC, CAE—that is, angles DAB and BAE, which are two right angles—are equal to the three angles of the triangle ABC. Therefore the three angles of a triangle are equal to two right angles. Such is the proof of the Pythagoreans. (*fr. 136*)



Eudemus and his school tell us that these things—the application of areas, their exceeding and their falling short—are ancient discoveries of the Pythagorean muse. (*fr. 137*)

This [*Elements* 1.23] too is a problem, and the credit for its discovery belongs rather to Oenopides, as Eudemus tells us. (*fr. 138*)

Archimedes wished to show that a circle is equal to a certain rectilineal area, a thing which had been of old investigated by illustrious philosophers. For it is evident that this is the problem concerning which Hippocrates of Chios and Antiphon, who carefully searched after it, invented the false reasonings which, I think, are well known to those who have looked into the *History of Geometry* of Eudemus and the *Keria* of Aristotle. (*fr. 139*)

[Thomas, *Selections illustrating the history of Greek mathematics*, vol. 1, p. 234-253] (*fr. 140*)

[Thomas, *Selections illustrating the history of Greek mathematics*, vol. 1, p. 284-289] (*fr. 141*)

..., it was nevertheless Theaetetus who distinguished the powers (i.e. the squares) which are commensurable in length, from those which are incommensurable (i.e. in length), and who divided the more generally known irrational lines according to the different means, assigning the medial line to geometry, the binomial to arithmetic, and the apotome to harmony, as is stated by Eudemus, the Peripatetic. (*fr. 141 I*)

History of Arithmetic

The fact that they demonstrated things to do with concords through the foundational numbers is shown by Eudemus in the first book of his *History of Arithmetic*, who writes as follows about the Pythagoreans, in these words: '<They said> that the ratios of the three concords, the fourth, the fifth and the octave, turn out in the first <numbers> to be contained in the <number> 9; for 2 and 3 and 4 make 9.' (*fr. 142*)

History of Astronomy

And Eudemus, in the *Astronomical Histories*, says that Thales foretold the eclipse of the sun, which took place at the time that the Medians and the Lydians fought, in the reign of Cyaxares the father of Astyages over the Medes, and of Alyattus the son of Croesus over the Lydians. (*fr. 143*)

[Thales] seems by some accounts to have been the first to study astronomy, the first to predict eclipses of the sun and to fix the solstices; so Eudemus in his *History of Astronomy*. (*fr. 144*)

Eudemus, in his books *On Astronomy*, tells that Oenopides found the first obliquity of the zodiac and recognized the existence of the great year. According to him, Thales was able to see that the eclipses of the sun and the returns of this star to the solstices do not always occur after the same lapse of time. Anaximander maintains that the earth is suspended in space and moves around the center of the world. Anaximenes has shown that the moon receives its light from the sun and the manner in which it is eclipsed. Others have added new discoveries to these: that the stars move around the immobile axis which passes through their poles, that the planets move around the axis perpendicular to the zodiac; and that the axis of the stars and that of the planets are separated from one another by the side of the pentadecagon, and consequently by an angle of 24 degrees. (*fr. 145*)

Eudemus recounts that Anaximander was the first to have given an account of their [the planets'] sizes and distances, and he credits the Pythagoreans with the first ordering of their position. (*fr. 146*)

Therefore since Plato <was attending to> the considerable association of the Sun and Moon and the fact that they proceed innately from the same cause, he has treated their procession into the cosmos as something that has been conjoined. In fact, he was not even the one who initiated this hypothesis, but it was instead Anaxagoras who first assumed this, as Eudemos relates the story. (*fr.* 147)

..., the <astronomers> were content to find out on what hypotheses the phenomena concerning the stars which are said to wander could be preserved by means of uniform, ordered, circular motions. And, as Eudemos recorded in the second book of his astronomical history (and Sosigenes took this over from Eudemos), Eudoxus of Cnidus is said to be the first of the Hellenes to have made use of such hypotheses, ... (*fr.* 148)

But Eudemos recounts briefly for the sake of which phenomena <Callippus> thought that these spheres had to be added. He says that <Callippus> says that since the times between the solstices and the equinoxes differ by as much as Euctemon and Meton thought, the three spheres for each <of sun and moon> were not sufficient to preserve the phenomena, obviously because of the apparent non-uniformity in their motions. And Eudemos also recounts briefly and clearly for what reason <Callippus> added one sphere in the case of each of the three planets Mars, Venus, and Mercury. (*fr.* 149)

This compilation is based on the list of fragments given in Vitrac, *Quand? Comment? Pourquoi les textes mathématiques grecs sont-ils parvenus en Occident?*, appendix 6. The translations are all taken from existing editions:

Fr. 134–138: G. R. Morrow, *Proclus: A Commentary on the First Book of Euclid's Elements*. Fr. 139: G. J. Allman, *Greek Geometry from Thales to Euclid*. Fr. 141 I: W. Thomson, *The Commentary of Pappus on Book X of Euclid's Elements*. Fr. 142: A. Barker, *Porphyry's Commentary on Ptolemy's Harmonics*. Fr. 143: W. Wilson, *Clement of Alexandria, Stromata*, in *Ante-Nicene Fathers*, vol. 2. Fr. 144: R. D. Hicks, *Diogenes Laertius, Lives of Eminent Philosophers*, vol. 1. Fr. 145: R. and D. Lawlor, *Theon of Smyrna, Mathematics Useful for Understanding Plato*. Fr. 146, 148–149: I. Mueller, *Simplicius: On Aristotle On the Heavens 2.10–14*. Fr. 147: D. Baltzly, *Proclus: Commentary on Plato's Timaeus*, vol. 5, book 4.

— Marius Kempe